

# Concepts Live in the Write Direction: Layered Evidence for Orthogonal, Composable LoRA Concept Spaces

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CHORA joint workspace — Experiment 10, same-day report

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## Abstract

Two low-rank adapters fine-tuned on disjoint domains (Spring Festival vs. summer) are shown to encode *separable concept spaces*: their weight updates  $\Delta W$  are near-orthogonal (global overlap  $-0.002$  to  $0.135$  across three seeds) while their input (read) subspaces are indistinguishable (all pairwise top- $k$  similarities  $\approx 3.8$ , flat). A data-mixture “intermediate” adapter is an *interpolation*, not a third space (registered null model wins). Behavioural (lexical) separation is strong (hit rate  $0.84$  vs  $0.11$ ), yet output distributions of the two specialists are *closer to each other than either is to the base* at every position of an answer, on content and style tokens alike — composability is therefore a property of the weight geometry plus the gate, not of output distinguishability. A learned gate ( $0.90$  accuracy; routed hit  $0.800$ ) beats a lexical router ( $0.80$ ;  $0.607$ ), and the composed runtime  $W = W_0 + \sum_i p_i \Delta W_i$  ships as 374–392 MB GGUF at 150–166 tokens/s on an M1 laptop. Cross-domain replication over five domains (adding code, legal, medical) keeps every specialist pair within  $|\Delta W_{\text{ov}}| < 0.13$ , with the interpolation anchor ( $0.62/0.66$ ) proving the metric detects non-orthogonality when it exists. All numbers are pinned, four-way hash-verified, and reproducible from committed scripts; one self-audited sampling bug and its corrected rerun are reported in full.

## 1 Setup

Base: Qwen2.5-0.5B (frozen, manifest-verified bytes). Adapters: LoRA  $r=16$ , 7 target modules, lr  $10^{-4}$ , 6 epochs, seeds 13/14/15; corpora of 55 (spring, summer), 110 (mixture), 40 (code/legal/medical, deviation registered) hand-written Chinese QA pairs. Probes: 40 questions (15/15/10), hashed *before* corpus authoring; 6-gram leakage tripwire = 0. Pre-registration precedes all runs (PREREG.md + four addenda); negative results are reported as first-class. Venue: `chora/experiments/exp10-concept-adapters`.

## 2 Method summary

**Behaviour (10a):** 5 sampled generations/probe; lexical hit rates; top-50 answer-start distributions, JS over union support. **Geometry (10b/10f):** per-module SVD of  $A \in \mathbb{R}^{r \times d_{\text{in}}}$ ,  $B \in \mathbb{R}^{d_{\text{out}} \times r}$ ; read-similarity  $\|V_i V_j^\top\|_F$  at 90%-energy  $k$ , write-similarity  $\|U_i^\top U_j\|_F$ ; basis-invariant global  $\langle \Delta W_i, \Delta W_j \rangle / (\|\Delta W_i\| \|\Delta W_j\|)$ ; random-flat null  $k/\sqrt{d}$ ; invasion = spectral band of the base weights the adapters write into. **Sequence (10c):** teacher-forced per-position JS on greedy reference texts; content-vs-style token classification under two definitions (function-word-inclusive, and punctuation-only). **Systems (10d/10g/GGUF):** multi-adapter runtime with

lexical router, then logistic gate on frozen-base sentence embeddings; merge, convert to GGUF f16, quantize q4\_K\_M, llama-bench tg128.

### 3 Results

| pair                           | read V-A ( $k90$ ) | write U-B ( $k90$ ) | $\Delta W_{ov}$ (3 seeds) |
|--------------------------------|--------------------|---------------------|---------------------------|
| spring–summer                  | 3.818              | 0.55–1.24           | −0.002...0.135            |
| spring–mid                     | 3.834              | 1.76–2.43           | 0.009...0.635             |
| summer–mid                     | 3.833              | 1.86–2.51           | 0.013...0.661             |
| code/legal/medical cross pairs | 3.785–3.790        | 1.07–1.16           | $\leq 0.129$              |

Table 1: Orthogonal writes, shared reads; the mixture interpolates.

**H1 fails with reversed sign** (JS spring–summer 0.417/0.380 < base–arm 0.551/0.575); it survives full-sequence averaging but shrinks  $3\times$  (gap 0.165  $\rightarrow$  0.060) and remains smallest on *both* content and style classes under either definition — the plan’s content/style rescue is disconfirmed. **H2 passes** (0.84/0.49 vs base 0.11/0.33; cross  $\leq 0.24$ ; neutral  $\approx 0$ ). **H4 fails; registered null wins** (mixture = interpolation). **H6 passes:** max  $|\Delta W_{ov}| = 0.1294$  across all 10 specialist pairs with positive control 0.618/0.661. **Invasion:** all adapters write the base’s *late* spectrum (band percentile  $\approx 0.71$ ; 1.5% energy in top-10% directions) — independent instrument, same finding as the workspace’s middle-eigenfunction thesis. **Systems:** gate 0.90 vs router 0.80 accuracy; routed hit 0.800 vs 0.607; q4\_K\_M 374–392 MB at  $149.6 \pm 11.9/154.6 \pm 9.1/165.8 \pm 4.1$  t/s (target was 20).

### 4 Self-audit

The first router run reseeded *inside* the sampling loop: five “independent” draws were identical (reported 0.70 was a single-draw number in five costumes). Disclosed the same day, fixed, rerun (v2: 0.607); v1 bytes remain pinned and immutable. Two further first-run bugs (temperature polarity, negative concentration floor) were exposed by the companion park-side breathing experiment (C3) and fixed in both artefacts the same day.

### 5 Reading

Concept separation is where the adapter *writes*; what it *reads* is the shared commonsense of the base.  $W_0 + \sum_i p_i \Delta W_i$  is therefore feasible as a composition law (near-orthogonality measured, five domains, one base size) *and* needs the gate: outputs alone do not reveal the concepts (10c), so the  $\sum$  must be carried by a learned selector (10g). A time-modulated prior pointer is the selector’s natural extension — previewed symbolically in the sister park the same day.

### 6 Limits and next

One 0.5B base, toy corpora, one gate class; magnitudes not production claims. Exp11: 5 bases  $\times$  4 domains + universal-LoRA probe; MLP/neural controller; perplexity accounting for merged bf16.

Bytes: artifacts/results/exp10/ (13 pinned files, manifest four-way 0 failures); scripts train\_lora.py · bench.py · analyze\_10a.py · svd\_10b.py · svd\_seedcheck.py · kl\_10c.py · kl\_10c\_grouped\_S1.py · router\_10d.py · gate\_10g.py · gate\_routed5.py · svd\_10f.py · gguf\_10e.sh; git: chora@a7a289e  $\rightarrow$  e0e609f (this day’s receipts).